HopeAI

AI-powered clinical development

- Integrating clinical insights with statistical innovations

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AI RUDDER

Clinical trial design: A lengthy, iterative process ready for AI acceleration

Clinical Evidence

1. Identify **unmet needs**
2. Define **patient population**
3. Choose **comparators**
4. Determine **benchmarks**

Statistical Considerations

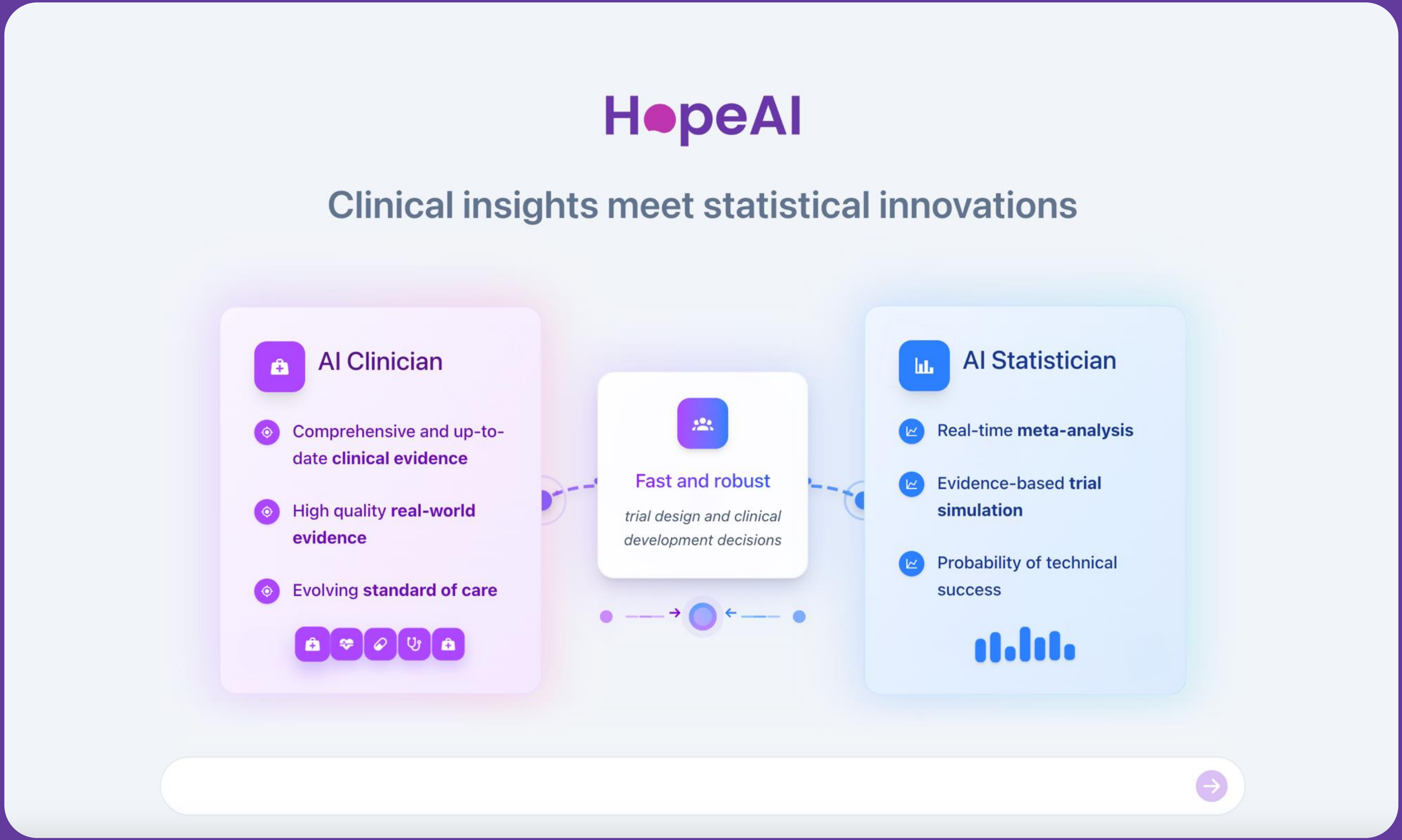
5. Evidence **synthesis**
6. **Sample size** determination
7. Patient **randomization**
8. **Endpoint** optimization

Clinical Team

Statistical Team

6-12 months of back-and-forth discussions, **$1M** in potential revenue delay **each day**

HopeAI brings clinical insights and statistical innovations to trial leaders’ fingertips



Outline

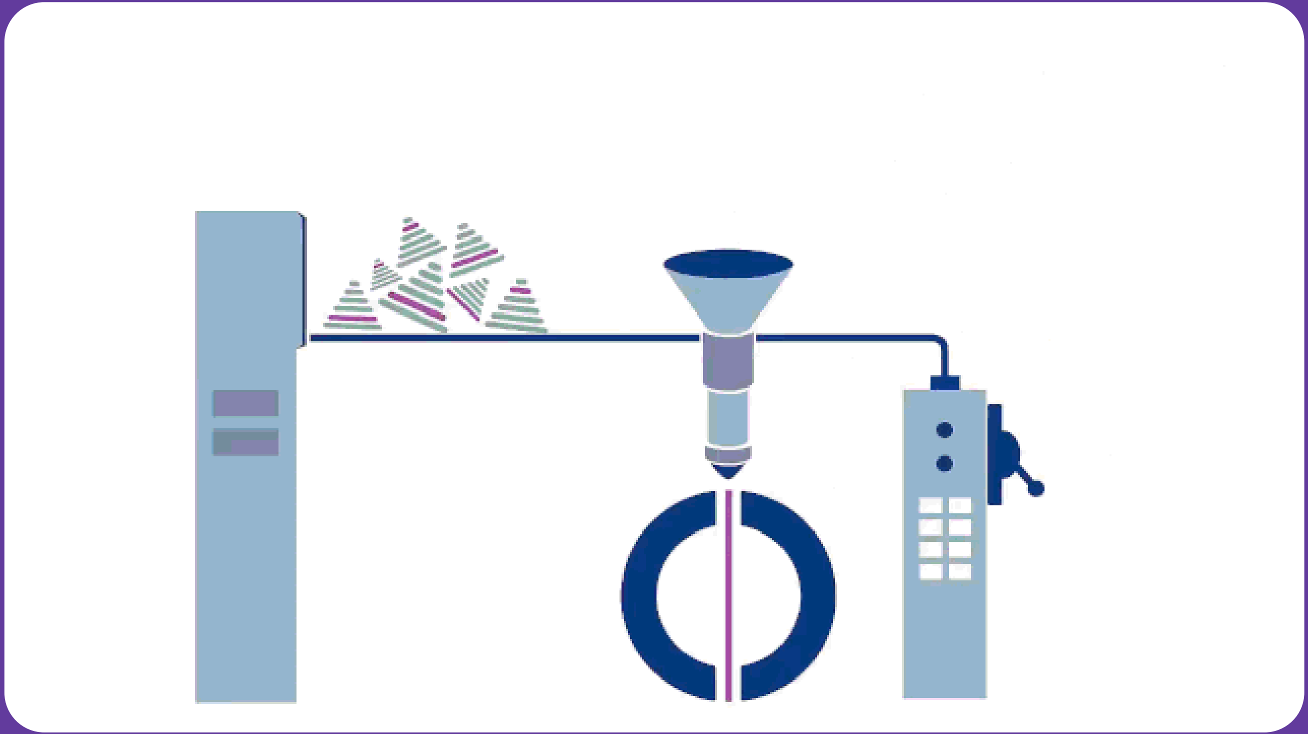
- **PURE Evidence** – Human-Curated Clinical Evidence
- **SynthIPD** – Synthetic Individual Patient Data
- **AI Teammates** – AI Clinician and AI Statistician

The significant cost of systematic literature review

5
co-workers

16
hours/week

6-16
months



Source: https://cccrg.cochrane.org/infographics

HopeAI enables live meta-analysis with PURE Evidence

Meta Analysis

✓ Choose Endpoint — Select analysis type
2 Select Data — Pick studies & filters
3 Report — View results

	NCTID	Analysis Population	Subgroup Type	Subgroup	Sample Size	mPFS	Unit	Lower CI
	NCT01650376	ITT	-	-	54.00	7.8	months	3.70
	NCT01650376	Subgroup	BIOMARKER	BRCA Mutated	23.00	12.1	months	9.40
	NCT01650376	Subgroup	BIOMARKER	BRCA Wild-type	24.00	4.8	months	2.30
	NCT01650376	Subgroup	TREATMENT_LINE	Prior Lines Treatment = 1-3	31.00	9.2	months	7.50
	NCT01650376	Subgroup	TREATMENT_LINE	Prior Lines Treatment≥4	23.00	3.7	months	2.60
	NCT01690598	ITT	-	-	27.00	2.8	months	2.60
	NCT02484404	ITT	-	-	17.00	16.1	months	4.50
	NCT02484404	Subgroup	BIOMARKER	DDR Positive	-	16.1	months	7.80
	NCT02484404	ITT	-	-	19.00	1.8	months	0.90
	NCT02502266	ITT	-	-	173.00	3.4	months	-
	NCT02502266	ITT	-	-	167.00	5.2	months	-
	NCT02502266	ITT	-	-	170.00	4	months	-
	NCT02502266	Subgroup	DISEASE_STATUS	Measurable Disease	140.00	-	-	-

Total 58 items 1 100 / page

← Back

Run Analysis →

AI workflow + expert curation for PURE Evidence

(Precise, Up-to-date, Reliable, Easily accessible)

Data Sources

Publications

Conferences

FDA/EMA Reviews

Drug Labels

Study Protocols

Stat Analysis Plans

→

Extraction

Agentic AI Workflow

- Publication selection agent
- Study population agent
- Specialized agents for tables, figures, paragraphs

AI + Human Curation

- Multi-AI cross validation
- Human consensus

99% accuracy

→

Standardization

Patient Segmentation

- I/E criteria
- Baseline characteristics
- Patient subgroups

Evidence Consolidation

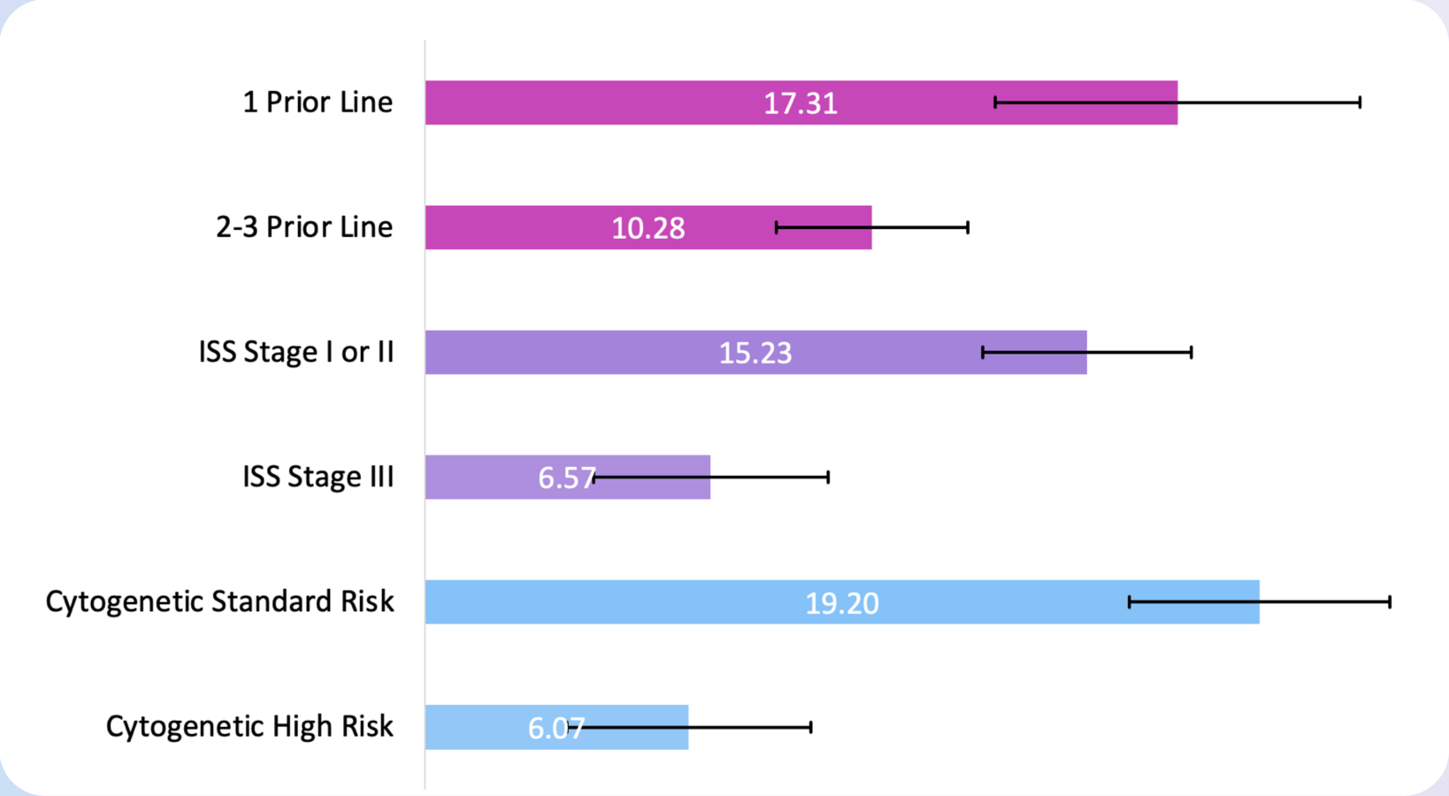
- Primary publications
- Secondary publications
- FDA adjudication

→

PURE Evidence

Use Case 1: Determine the right sample size based on full clinical evidence

Pooled median PFS of early line RRMM by baseline characteristics



80% of clinical trials were designed without conducting systematic review, due to tight timeline and lack of reliable solutions.

Small sample size leads to underpowered study.

Large sample size leads to

- Longer duration of study
- Late to market
- Poor return on investment

Median PFS of early line len-exposed/refractory MM by baseline characteristics

	APPOLO (DPd)	CARTITUDE-4 (DPd/PVd)	OPTIMISMM (PVd)	CANDOR (Kd)	ENDEAVOR (Kd)	ARROW (Kd)
1 Prior Line	14.1	17.4	17.84	21.3	15.6	
2-3 Prior Line	10.7	10.2		12.5	9.7	12.1/8.9
ISS Stage I or II	19.3/12.3	14.7		15.8		
ISS Stage III	6.1	6.2		7.4		
Cytogenetic High Risk	5.8	6.7	8.44	5.6	8.8	
Cytogenetic Standard Risk	21	20.6		16.6		

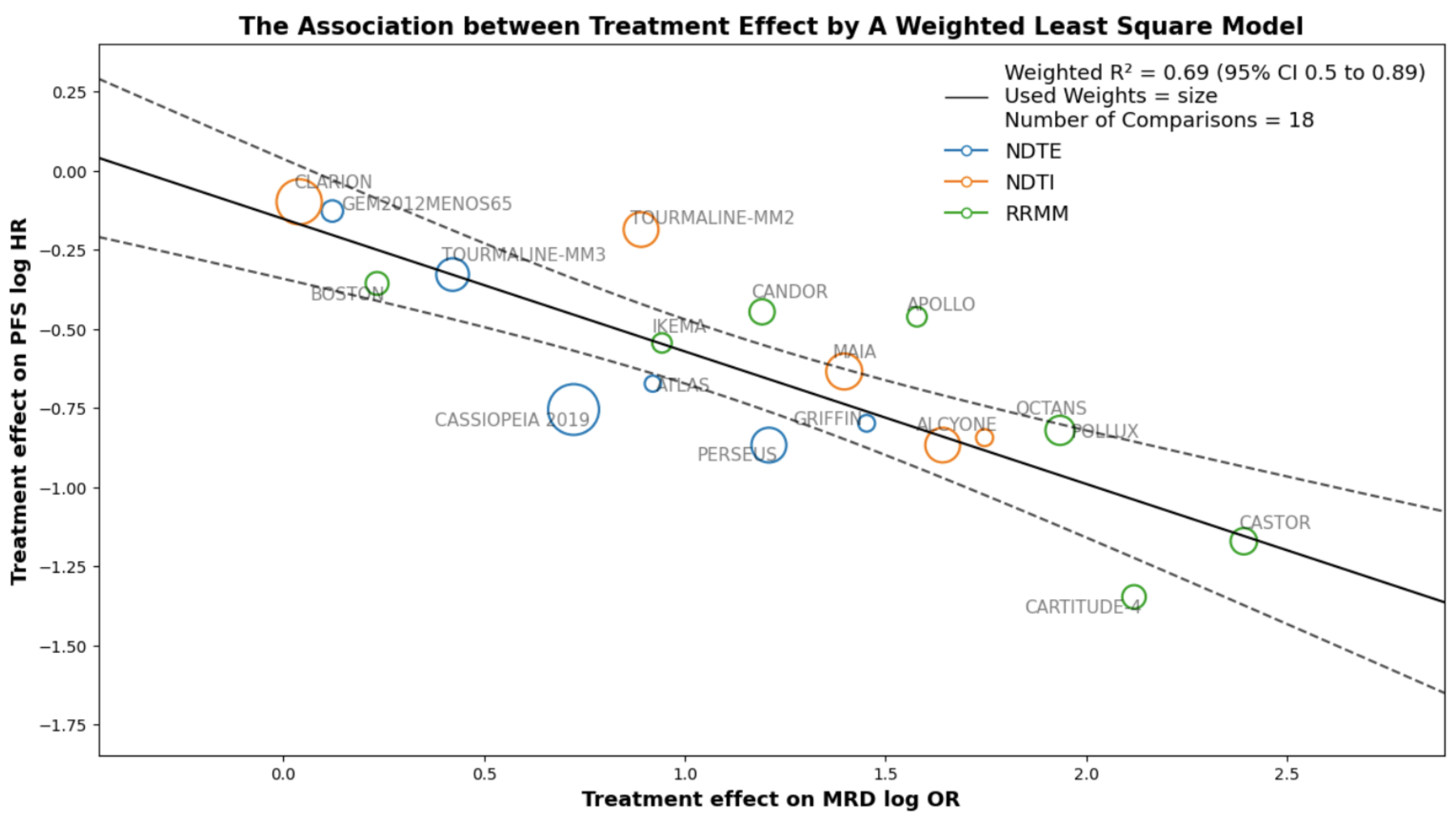
- Clinical trial outcomes by subgroups for Kd, DPd, PVd and EPd for early line lenalidomide-exposed/refractory RRMM were included.
- Data was extracted by AI and curated by human.

Use Case 2: Address FDA’s comments on a single arm study in AML with comprehensive evidence

Complete remission (CR) rate of non-targeted therapies for R/R AML patients

Study ID	Study Name	Agent	Population	Sample Size	Median Age (years)	Age Range (years)	CR Rate	CRi Rate	Median OS (months)
NCT02607059	PETHEMA	Venetoclax + HMA	R/R AML	51	68	25-82	10.40%	2.00%	3.4
NCT02421939	ADMIRAL	Salvage chemotherapy	FLT3-mutated R/R AML	124	62	19-85	10.50%	11.30%	5.6
NCT03182244	COMMODORE	Salvage chemotherapy	FLT3-mutated R/R AML (Asian)	118	50		10.20%	9.30%	5
NCT01147939	CLAVELA	Elacytarabine	R/R AML	191	62	22-89	15.00%	8.00%	3.5
NCT01147939	CLAVELA	Investigator Choice	R/R AML	190	63	19-83	12.00%	9.00%	3.3
NCT02152956		Flotetuzumab	R/R AML	50	64	27-82	12.00%		3.2

Use Case 3: Validated surrogacy of MRD negativity in multiple myeloma based on SynthIPD



In partnership with:

MAYO CLINIC

Fred Hutch Cancer Center

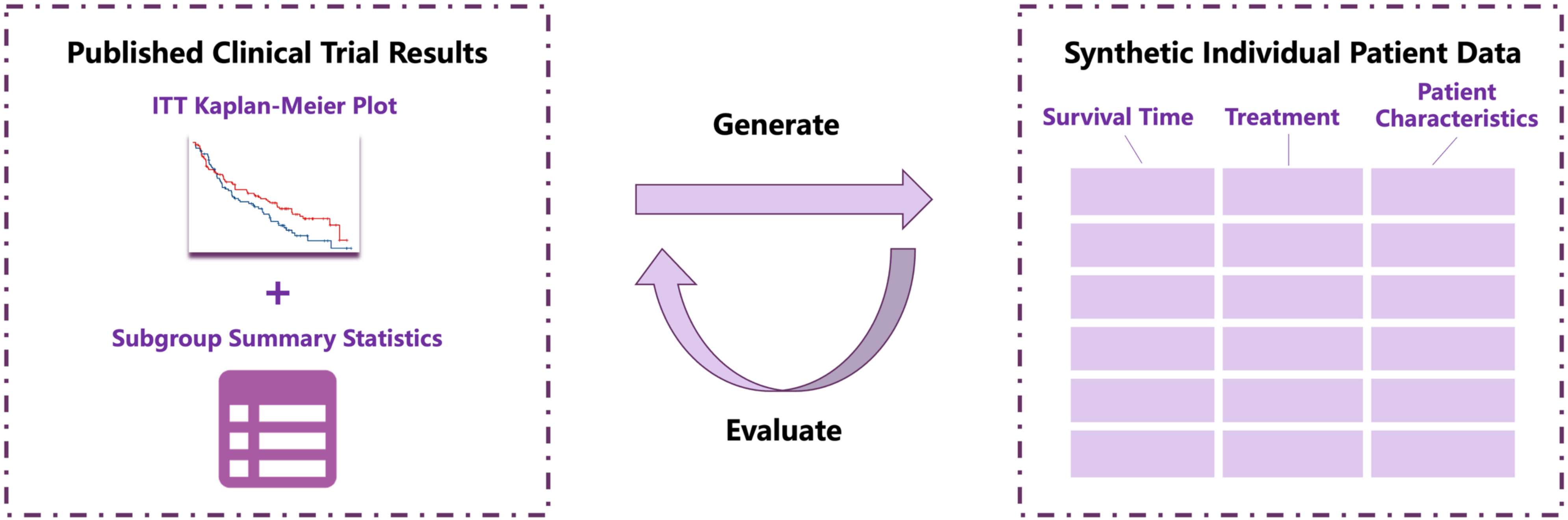
Featured at:

International Myeloma Society

ASCO
AMERICAN SOCIETY OF CLINICAL ONCOLOGY

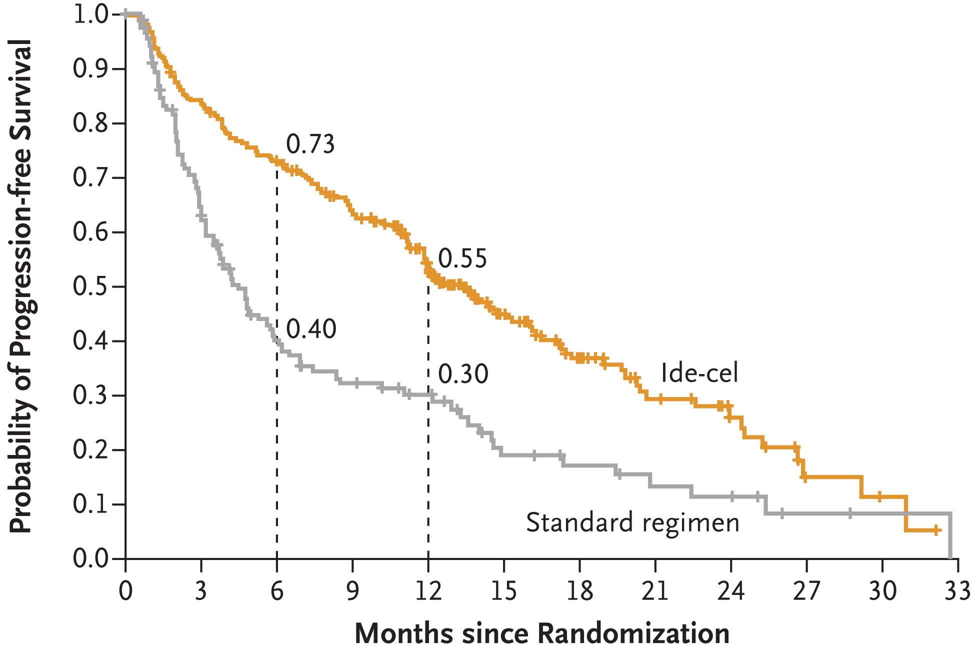
SynthIPD – Synthetic Individual Patient Data

Reconstructing survival outcome and patient covariate from KM plots and subgroup summary statistics



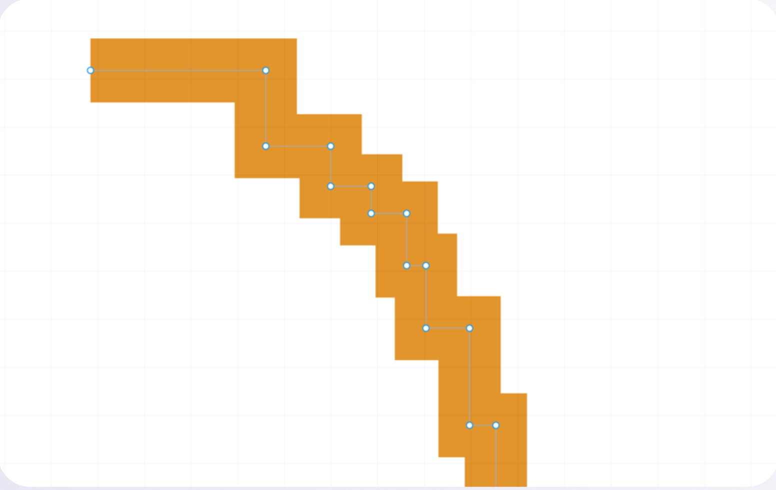
Digitization of the KM plot at pixel-level accuracy

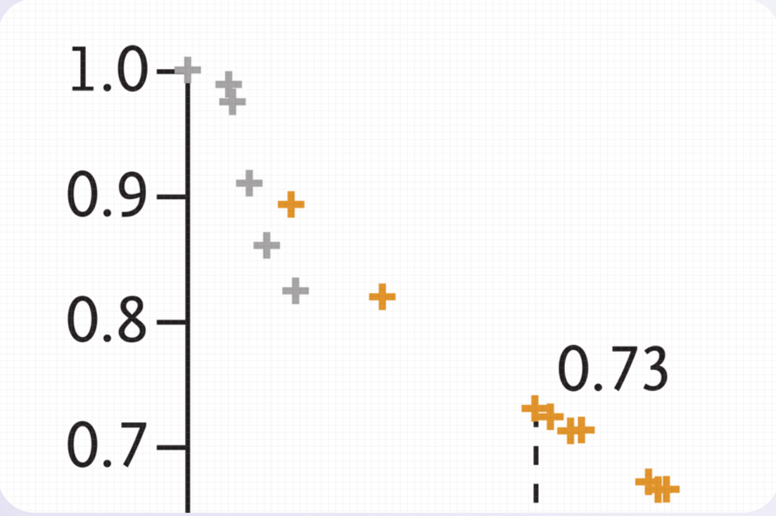
Enables reconstruction of the exact survival time and censor status of each patient in the ITT population



No. at Risk

Ide-cel	254	206	178	149	110	62	40	22	14	4	2	0
Standard regimen	132	75	42	32	25	13	10	7	6	2	1	0





Simulation of covariates to match the subgroup results

A novel iterative algorithm with substantial computing power was used to simulate patient characteristics

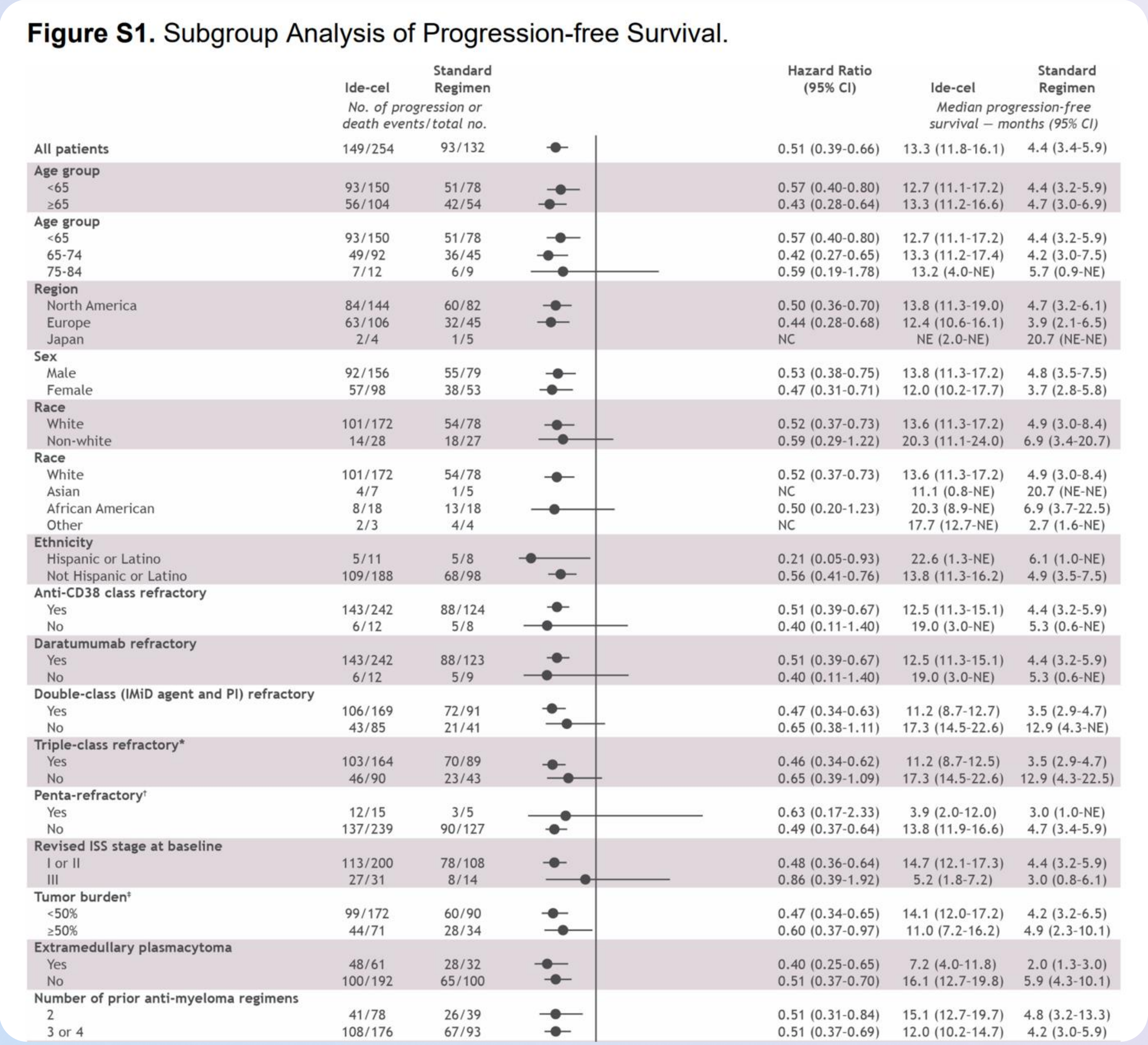
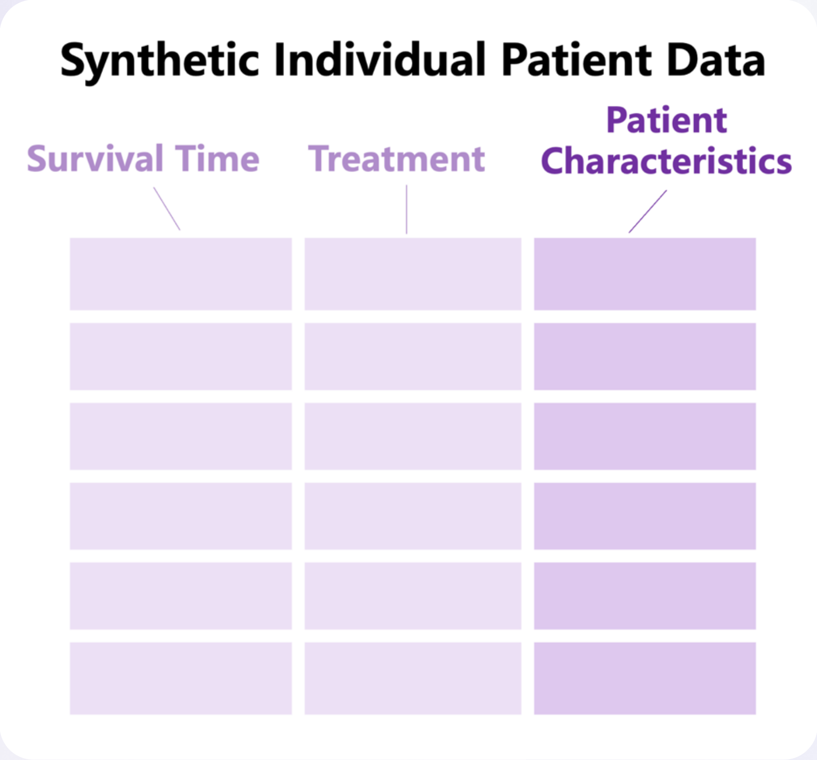
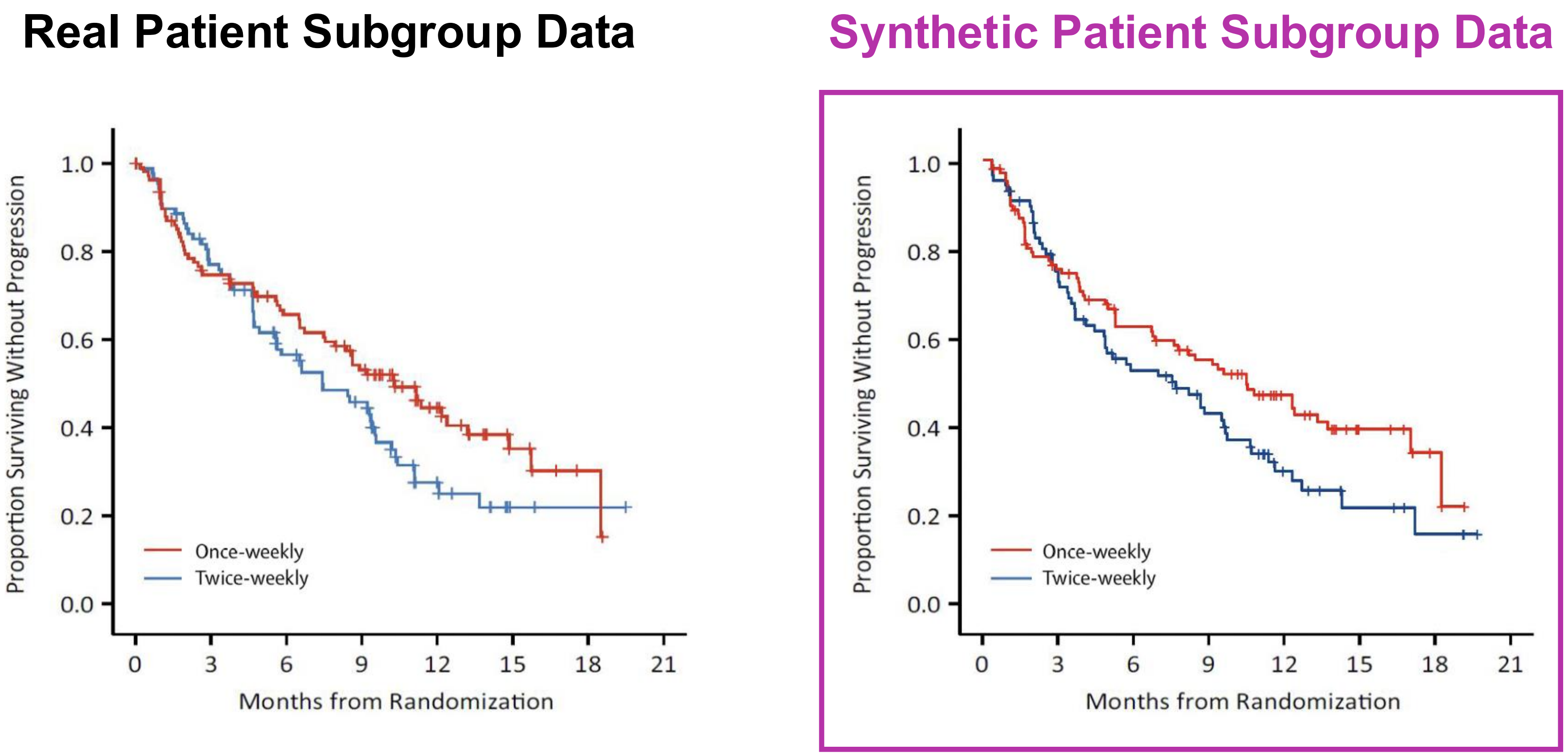


Figure S1. Subgroup Analysis of Progression-free Survival.

	Ide-cel	Standard Regimen	Hazard Ratio (95% CI)	Ide-cel	Standard Regimen
	No. of progression or death events/total no.			*Median progression-free survival — months (95% CI)*	
All patients	149/254	93/132	0.51 (0.39-0.66)	13.3 (11.8-16.1)	4.4 (3.4-5.9)
Age group					
<65	93/150	51/78	0.57 (0.40-0.80)	12.7 (11.1-17.2)	4.4 (3.2-5.9)
≥65	56/104	42/54	0.43 (0.28-0.64)	13.3 (11.2-16.6)	4.7 (3.0-6.9)
Age group					
<65	93/150	51/78	0.57 (0.40-0.80)	12.7 (11.1-17.2)	4.4 (3.2-5.9)
65-74	49/92	36/45	0.42 (0.27-0.65)	13.3 (11.2-17.4)	4.2 (3.0-7.5)
75-84	7/12	6/9	0.59 (0.19-1.78)	13.2 (4.0-NE)	5.7 (0.9-NE)
Region					
North America	84/144	60/82	0.50 (0.36-0.70)	13.8 (11.3-19.0)	4.7 (3.2-6.1)
Europe	63/106	32/45	0.44 (0.28-0.68)	12.4 (10.6-16.1)	3.9 (2.1-6.5)
Japan	2/4	1/5	NC	NE (2.0-NE)	20.7 (NE-NE)
Sex					
Male	92/156	55/79	0.53 (0.38-0.75)	13.8 (11.3-17.2)	4.8 (3.5-7.5)
Female	57/98	38/53	0.47 (0.31-0.71)	12.0 (10.2-17.7)	3.7 (2.8-5.8)
Race					
White	101/172	54/78	0.52 (0.37-0.73)	13.6 (11.3-17.2)	4.9 (3.0-8.4)
Non-white	14/28	18/27	0.59 (0.29-1.22)	20.3 (11.1-24.0)	6.9 (3.4-20.7)
Race					
White	101/172	54/78	0.52 (0.37-0.73)	13.6 (11.3-17.2)	4.9 (3.0-8.4)
Asian	4/7	1/5	NC	11.1 (0.8-NE)	20.7 (NE-NE)
African American	8/18	13/18	0.50 (0.20-1.23)	20.3 (8.9-NE)	6.9 (3.7-22.5)
Other	2/3	4/4	NC	17.7 (12.7-NE)	2.7 (1.6-NE)
Ethnicity					
Hispanic or Latino	5/11	5/8	0.21 (0.05-0.93)	22.6 (1.3-NE)	6.1 (1.0-NE)
Not Hispanic or Latino	109/188	68/98	0.56 (0.41-0.76)	13.8 (11.3-16.2)	4.9 (3.5-7.5)
Anti-CD38 class refractory					
Yes	143/242	88/124	0.51 (0.39-0.67)	12.5 (11.3-15.1)	4.4 (3.2-5.9)
No	6/12	5/8	0.40 (0.11-1.40)	19.0 (3.0-NE)	5.3 (0.6-NE)
Daratumumab refractory					
Yes	143/242	88/123	0.51 (0.39-0.67)	12.5 (11.3-15.1)	4.4 (3.2-5.9)
No	6/12	5/9	0.40 (0.11-1.40)	19.0 (3.0-NE)	5.3 (0.6-NE)
Double-class (IMiD agent and PI) refractory					
Yes	106/169	72/91	0.47 (0.34-0.63)	11.2 (8.7-12.7)	3.5 (2.9-4.7)
No	43/85	21/41	0.65 (0.38-1.11)	17.3 (14.5-22.6)	12.9 (4.3-NE)
Triple-class refractory*					
Yes	103/164	70/89	0.46 (0.34-0.62)	11.2 (8.7-12.5)	3.5 (2.9-4.7)
No	46/90	23/43	0.65 (0.39-1.09)	17.3 (14.5-22.6)	12.9 (4.3-22.5)
Penta-refractory†					
Yes	12/15	3/5	0.63 (0.17-2.33)	3.9 (2.0-12.0)	3.0 (1.0-NE)
No	137/239	90/127	0.49 (0.37-0.64)	13.8 (11.9-16.6)	4.7 (3.4-5.9)
Revised ISS stage at baseline					
I or II	113/200	78/108	0.48 (0.36-0.64)	14.7 (12.1-17.3)	4.4 (3.2-5.9)
III	27/31	8/14	0.86 (0.39-1.92)	5.2 (1.8-7.2)	3.0 (0.8-6.1)
Tumor burden‡					
<50%	99/172	60/90	0.47 (0.34-0.65)	14.1 (12.0-17.2)	4.2 (3.2-6.5)
≥50%	44/71	28/34	0.60 (0.37-0.97)	11.0 (7.2-16.2)	4.9 (2.3-10.1)
Extramedullary plasmacytoma					
Yes	48/61	28/32	0.40 (0.25-0.65)	7.2 (4.0-11.8)	2.0 (1.3-3.0)
No	100/192	65/100	0.51 (0.37-0.70)	16.1 (12.7-19.8)	5.9 (4.3-10.1)
Number of prior anti-myeloma regimens					
2	41/78	26/39	0.51 (0.31-0.84)	15.1 (12.7-19.7)	4.8 (3.2-13.3)
3 or 4	108/176	67/93	0.51 (0.37-0.69)	12.0 (10.2-14.7)	4.2 (3.0-5.9)



SynthIPD vs real IPD on patient subgroup of interest



Kaplan-Meier curves for patients who are refractory to Bortezomib

Note: the subgroup KM plot of real IPD is displayed for comparison purpose, and was not used in generating the SynthIPD

Technical paper of SynthIPD is now available on arXiv

arXiv > stat > arXiv:2509.16466

Statistics > Applications

[Submitted on 19 Sep 2025]

SynthIPD: assumption-lean synthetic individual patient data generation

Zixuan Zhao, Zexin Ren, Guannan Zhai, Feifang Hu, Will Ma, En Xie, Qian Shi

Individual patient data (IPD) are essential for statistical inference in clinical research. However, privacy concerns, high data-sharing costs, and restrictive access often make IPD unavailable. Conventional synthetic data generation usually relies on black box models such as generative adversial networks. These methods, however, requires a large piece of IPD for model training, may be ungeneralizable and lacks interpretability. This paper introduces an assumption-lean, three-step methodology for generating synthetic IPD with survival endpoints only based on published clinical trial articles. The method mainly leverages Kaplan-Meier (KM) curves with at-risk/censoring information and subgroup-level summary statistics. It digitizes the KM curve using Scalable Vector Graphics (SVG) beyond pixel accuracy and then generates synthetic covariates based on the statistics. We illustrate the method's potential through 2 detailed case studies and simulation studies. The method offers important implications, enabling high-fidelity IPD generation to support evidence-based medical decisions.

Subjects: **Applications (stat.AP)**
Cite as: arXiv:2509.16466 **[stat.AP]**
(or arXiv:2509.16466v1 **[stat.AP]** for this version)
https://doi.org/10.48550/arXiv.2509.16466

Our AI Clinician outperforms leading LLMs by 2x in accuracy and comprehensiveness of clinical evidence

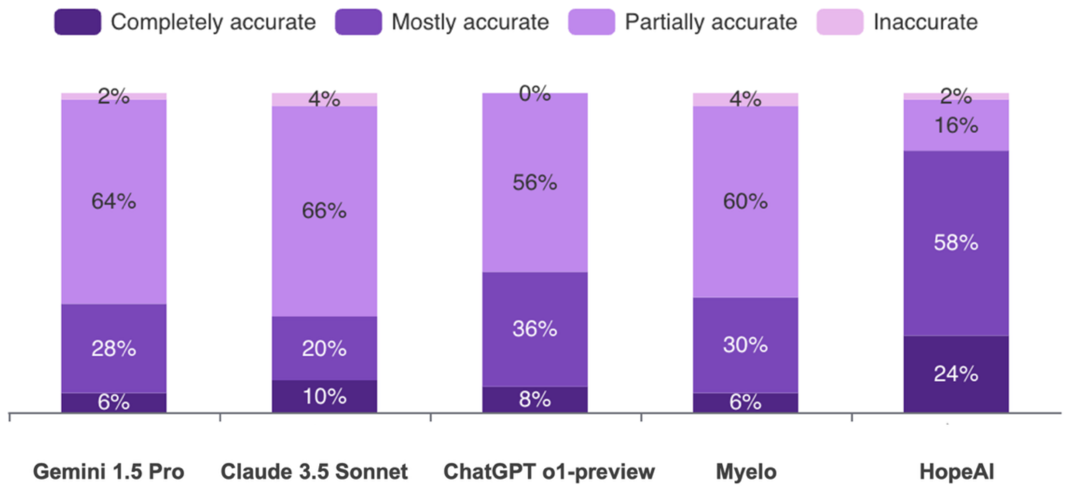
According to a large scale (4,500 data points) blinded evaluation by physicians from:

MAYO CLINIC

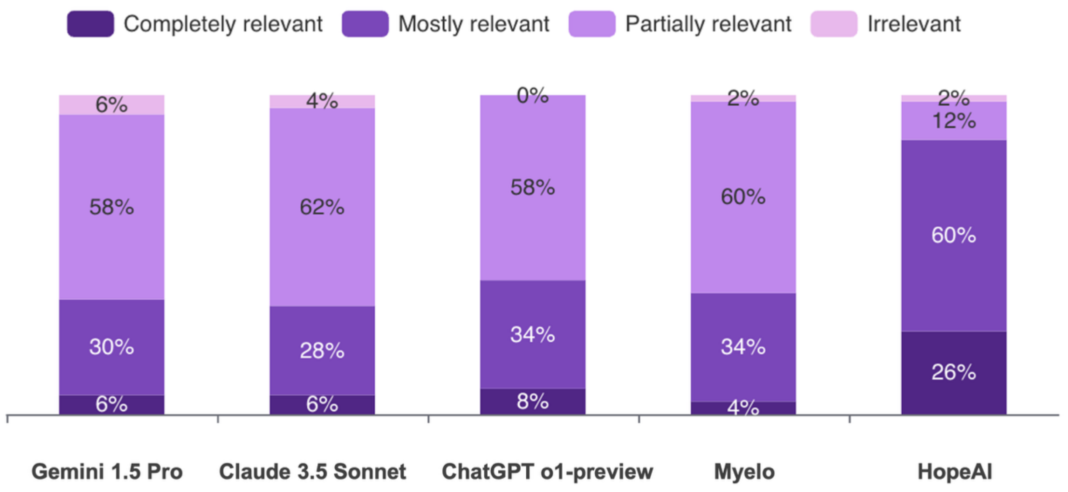
Fred Hutch Cancer Center

HemOnc.org

Is the answer accurate?



Is the answer relevant and up-to-date?



Our AI Statistician leverages decades of statistical innovations in trial design and inferences

100+

research papers

2

FDA whitepapers

Journal of the American Statistical Association ›

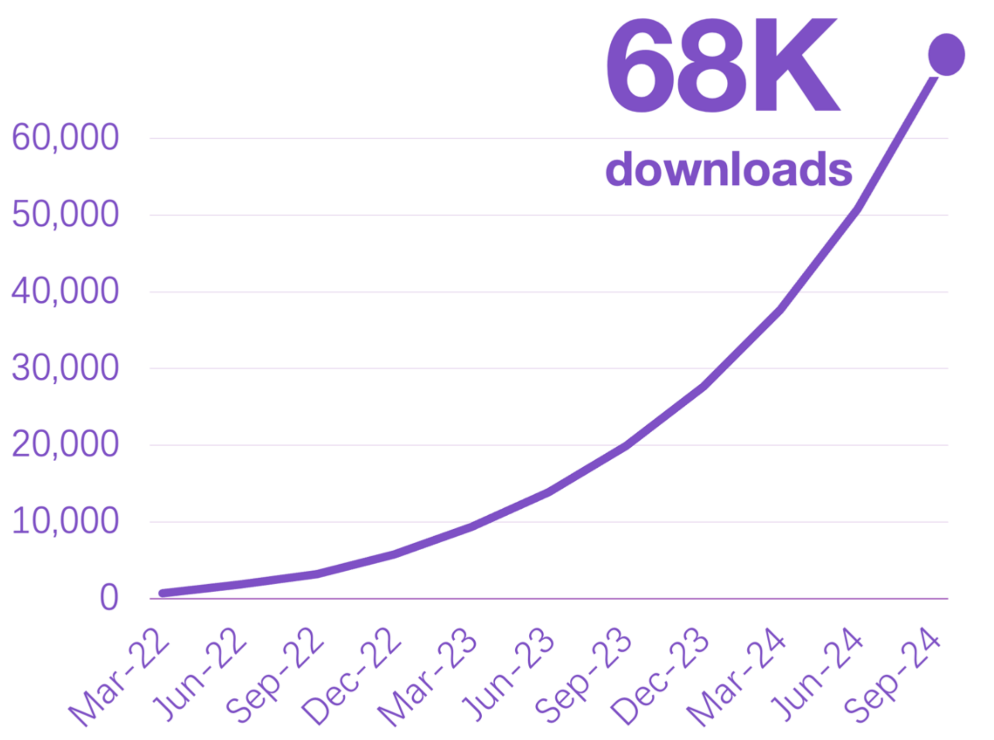
Volume 119, 2024 - Issue 545

1,178	3	1
Views	CrossRef citations to date	Altmetric

Theory and Methods

A New and Unified Family of Covariate Adaptive Randomization Procedures and Their Properties

Wei Ma, Ping Li, Li-Xin Zhang & Feifang Hu



Downloads of CARAT Adaptive Design Package

The path to AI-powered clinical development

Stage 1: Data and Tools

Build the data foundation and analytical tools for clinical development teams

Stage 2: AI Teammates

Create AI teammates to collaborate within the clinical development team

Stage 3: AI-powered Clinical Development

Implement end-to-end autonomous clinical development under human supervision

HopeAI

Bring hope to patients through AI.

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